

Species-Aware Module Review: MnmA tRNA Wobble-Uridine 2-Thiolation Sulfur Relay in *Pseudomonas putida* KT2440

Target: *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP0000000556)

Pathway bucket: KEGG ppu04122 "Sulfur relay system" **Module focus:** Cytoplasmic sulfur relay producing 2-thiouridine (s²U34) at the tRNA wobble position (the thiolation half of mnm⁵s²U34) **Curation date:** 2026-08-11 · Iterations 1–5 (final)

1. Executive Summary

The MnmA 2-thiouridine relay is **fully encoded and should be marked COVERED** in *P. putida* KT2440. Every biochemical step of the experimentally reconstituted *E. coli* relay — IscS → TusA

→ TusBCD → TusE → MnmA → s²U34

(

P 16387657

) — has a clear ortholog in the KT2440 proteome, and the four central mediators sit in a **syntenic** **tusD-tusC-tusB-tusE** **operon (PP_3993–PP_3996)**. Because both organisms are gammaproteobacteria and the *tus* genes are co-localized, transfer of the *E. coli* mechanism to KT2440 is **strong**.

The main curation issue is **not gaps but over-scope**: KEGG map04122 lumps three chemically distinct, IscS-fed sulfur pathways (2-thiouridine relay, molybdopterin/Moco biosynthesis, thiamine/4-thiouridine) into one "Sulfur relay system" bucket. Only **7 of the 22 candidate genes** belong to the 2-thiouridine module. The initial **paralog ambiguity** at the entry points was **resolved in Iteration 2** by sequence identity + InterPro signature + operon context: **IscS** = **PP_0842** (isc operon, IscS-specific domain) and **TusA** = **PP_2116** (specific Sulphur_carrier_TusA signature) are the relay-committed genes, while PP_2435 and PP_1233 are divergent accessory paralogs. A minor residual issue is **generic EC 2.8.1.- / unnamed** annotations on some Tus subunits (PP_3994/PP_3995).

Evidence is **homology- and synteny-based**; there is, to my knowledge, **no direct KT2440-specific biochemical or genetic characterization** of this relay. Confidence rests on the strength of the *E. coli* model plus KT2440 orthology/synteny.

2. Target-Organism Pathway Definition

Included biochemical process. ATP-dependent 2-thiolation of the wobble uridine (U34 → s²U34) in tRNA^{Glu}, tRNA^{Lys}, tRNA^{Gln}, catalyzed by the thiouridylase **MnmA (EC 2.8.1.13)**, using sulfur delivered through a protein-only persulfide relay originating from the cysteine desulfurase **IscS (EC 2.8.1.7)**. s²U34 restricts codon–anticodon wobble and is required for accurate decoding of NNR (AAR/CAR/GAR) codons and for recognition by cognate aminoacyl-tRNA synthetases

(
P 16871210
),

P 26791911

Neighboring processes to keep SEPARATE: - **mn^{m5} side chain (MnmE/MnmG/GidA)**. MnmA installs only the 2-thio group; the 5-methylaminomethyl group is added by MnmE (PP: P0A175, 456 aa) and MnmG (Q88RW8, 631 aa), **both present in KT2440** — so the strain makes the complete mn^{m5}s²U34. These add carbon/nitrogen (not sulfur) and are outside the sulfur-relay bucket and this module (though adjacent — they finish the same nucleoside). - **Molybdopterin / Moco biosynthesis** (MoaA/B/C/D/E, MoeB). A separate IscS-fed sulfur pathway that KEGG map04122 also contains. - **Thiamine biosynthesis & 4-thiouridine (s⁴U8)** (ThiI, ThiS/ThiF). ThiI makes s⁴U8 at position 8 (not the wobble position) and feeds thiamine; distinct from MnmA. - **Fe-S cluster assembly (Isc/Suf)**. Shares IscS as donor but is a different destination.

Alternate names / database definitions. The *E. coli* Tus genes were originally *yhhP/yheL/yheM/yheN/yccK* and renamed *tusA/tusB/tusC/tusD/tusE*

(
P 16387657
).

MnmA is also historically **TrmU / Asm / MnmA thiouridylase**. KEGG module context for this branch corresponds to the "2-thiouridine" portion of map00790/map04122; it is **not** a standalone KEGG MODULE, which is why no module YAML resolved.

3. Expected Step Model (reconstituted *E. coli* relay)

Step	Reaction / role	<i>E. coli</i> protein	Expected in KT2440?
S0	Cys → Ala + persulfide sulfur (desulfuration)	IscS (EC 2.8.1.7)	Yes (master donor)
S1	Accept persulfide from IscS; committed carrier	TusA	Yes
S2	Sulfur relay via TusD Cys78 persulfide (heterohexamer)	TusBCD complex	Yes
S3	DsrC-like mediator; stimulates TusA→TusD transfer; docks MnmA·tRNA	TusE	Yes
S4	ATP-dependent adenylation of U34 + sulfur insertion → s ² U34	MnmA (EC 2.8.1.13)	Yes (terminal enzyme)

Directionality

([P 16387657](#)):

IscS desulfurase activity is stimulated by TusA binding; IscS→TusA persulfide transfer; TusE stimulates TusA→TusD transfer and interacts with the MnmA–tRNA complex; TusD Cys78 is the catalytic sulfur-carrying residue

([P 16472754](#)):

MnmA traps an adenylated RNA intermediate before sulfur incorporation

([P 16871210](#)).

4. Candidate Genes and Evidence

4A. Core module — high confidence (mark COVERED)

Gene	Locus / UniProt	Length	Role	Evidence & caveats
mnmA	PP_4014 / Q88FR9	374 aa	Terminal thiouridylase, EC 2.8.1.13; UniProt "Catalyzes 2-thiolation of U34 → s ² U34"	Strong: 70.1% id to E. coli MnmA (P25745) ; conserves N-type PP-loop SGGVDS (adenylation) and both catalytic cysteines (Cys107/Cys203 ≡ EcCys101/Cys198). Promote to fetch-gene as module anchor.
tusD	PP_3993 / Q88FT9	130 aa	TusBCD catalytic subunit (Cys78 persulfide)	Best hit EcTusD 50.8% ; length ≈ EcTusD (128 aa). Catalytic Cys78-equivalent conserved (...AVVCIAAALRRG...). Catalytically essential subunit (P 16472754).
PP_3994	PP_3994 / Q88FT8	119 aa	TusC (metadata "TusC-like")	Best hit EcTusC 44.5% ; length = EcTusC (119 aa); in operon. Caveat: no UniProt gene symbol ; metadata protein name ("tRNA 5-MAM-2-thiouridine synthase, TusC-like") is a mis-descriptor — it is a TusBCD structural subunit, not the synthase.
PP_3995	PP_3995 / Q88FT7	98 aa	TusB (UniProt "Sulfurtransferase complex subunit TusB")	Best hit EcTusB 42.1% ; length ≈ EcTusB (95 aa); in operon. No gene symbol assigned.
tusE	PP_3996 / Q88FT6	111 aa	TusE / DsrC-like mediator, EC 2.8.1.-	Named <i>tusE</i> ; length ≈ EcTusE (109 aa); UniProt "Part of a sulfur-relay system." DsrC-family C-terminal Cys conserved (...PKPTNCI vs EcTusE ... PKPVKCI). Caveat: broad EC 2.8.1.-.
tusA	PP_2116 / Q88L21	80 aa	Sulfur carrier accepting IscS persulfide	Named <i>tusA</i> ; 57.5% id to EcTusA (P0A890) ; carries the specific IPR022931 (Sulphur_carrier_TusA) ; CxxP motif GLNCPEP. Relay carrier confirmed over paralog PP_1233 (§5).

Gene	Locus / UniProt	Length	Role	Evidence & caveats
iscS	PP_0842 / Q88PK8	404 aa	Master cysteine desulfurase, EC 2.8.1.7	72.5% id to EcIscS (P0A6B7) ; IscS-specific IPR010240 ; in the <i>isc</i> operon (PP_0842–0845). Relay donor confirmed over paralog PP_2435 (§5). Also serves Fe-S/Moco.

The **PP_3993→PP_3996 consecutive locus tags form a `tusD-tusC-tusB-tusE` operon**, a strong independent line of evidence that these four are a functional unit dedicated to the relay.

4B. Candidate genes that belong to OTHER sulfur sub-pathways (keep out of this module)

- **Molybdopterin/Moco:** moeB (PP_0735), moaC (PP_1292), moaD (PP_1293), moaE (PP_1294), PP_1969, moaB-I (PP_2122), PP_2482, moaA (PP_4597), moaB-II (PP_4600). → assign to a **Moco biosynthesis module**; `not_expected_in_target_taxon` for this 2-thiouridine module.
- **Thiamine / 4-thiouridine:** thiI (PP_5045, EC 2.8.1.4 → s⁴U8 + ThiS thiocarboxylate; UniProt PATHWAY "thiamine diphosphate biosynthesis"), thiS (PP_5105). → distinct module.
- **Generic rhodanases:** sseA (PP_5118, 3-mercaptopyruvate sulfurtransferase EC 2.8.1.2), rhdA (PP_4907). Non-specific single-domain sulfurtransferases; **not** core relay members (possible ancillary sulfur pools only).

5. Gaps, Ambiguities, and Likely Over-Annotations

- **No true gaps.** All five relay activities plus the desulfurase are encoded. The module is satisfiable.
- **Sulfur-donor paralogy — RESOLVED (Iteration 2).** Two cysteine desulfurases: **IscS (PP_0842)** vs **IscS-II (PP_2435)** (both EC 2.8.1.7, ~404–405 aa). Three concordant criteria identify **PP_0842 as the relay donor**: 72.5% identity to *E. coli* IscS (P0A6B7) vs 52.7% for PP_2435; only PP_0842 carries the IscS-specific InterPro **IPR010240 (Cys_deSase_IscS)**; and PP_0842 sits in the canonical **isc operon** (PP_0842 *iscS* – PP_0843 *iscU* – PP_0844 *iscA* – PP_0845 *hscB*). PP_2435 lacks the IscS signature and is flanked by a Xre/Res toxin–antitoxin

pair and an acyl-CoA dehydrogenase (no *isc/suf/csd* context) → **divergent accessory desulfurase, not the primary relay donor.**

- **Carrier paralogy — RESOLVED (Iteration 2).** Two TusA-family carriers: **TusA (PP_2116)** vs **TusA-I (PP_1233)**. **PP_2116 is the relay carrier:** 57.5% identity to *E. coli* TusA (P0A890) and it uniquely carries the specific InterPro **IPR022931 (Sulphur_carrier_TusA)**. PP_1233 is only 44.3% identical and carries only the generic **IPR001455 (TusA-like)** signature → likely a paralog serving a different sulfur branch (Moco/DsrE2-YeeD-type), consistent with *E. coli* TusA's multifunctionality

(

P 33526615

).

Both retain the catalytic CxxP persulfide motif (PP_2116 GLNCPEP, PP_1233 GLNCPLP), so both are functional sulfurtransferases; the discriminator is the family-specific InterPro signature, not the active site.

- **Over-propagation / naming issues.** PP_3994 protein name ("tRNA 5-methylaminomethyl-2-thiouridine synthase, TusC-like") over-claims enzymatic identity — it is a **TusBCD structural subunit (TusC)**, not the synthase (MnmA is). PP_3994/PP_3995 lack gene symbols; TusE and TusA-I carry broad "Sulfurtransferase (EC 2.8.1.-)" terms. These are annotation-transfer artifacts, not functional claims.
- **MnmA is single-copy.** Only one 2-thiouridylase (PP_4014) exists in the proteome, so the terminal-enzyme assignment is unambiguous (no paralog competition).
- **PP_1233 is low-evidence.** It carries only generic GO:0016740 (transferase activity) and UniProt annotationScore 1.0 (lowest tier); keep it excluded from the relay gene set pending characterization.
- **Species-transfer caveat (important).** All functional calls are **inferred from *E. coli*** (orthology + operon synteny + catalytic-residue conservation), **not from KT2440 experiments**. A targeted PubMed search (Iteration 4–5) found **no direct experimental characterization** of the MnmA/Tus relay in *P. putida* KT2440 or in *Pseudomonas* generally. Transfer is **strong** for gammaproteobacteria (short phylogenetic distance, conserved operon, all catalytic residues intact) but must be labeled inferred, not direct.

6. Module and GO-Curation Recommendations

Module step	Status	KT2440 gene(s)
Cysteine desulfurase (S source)	covered (PP_0842 high-confidence; PP_2435 = accessory, not relay)	IscS PP_0842
TusA sulfur carrier	covered (PP_2116 high-confidence; PP_1233 = other branch)	TusA PP_2116
TusBCD complex	covered	TusB PP_3995, TusC PP_3994, TusD PP_3993
TusE mediator	covered	TusE PP_3996
MnmA thiouridylase (s ² U34)	covered	MnmA PP_4014
Moco / thiamine / rhodanese genes	not_expected_in_target_taxon (<i>for this module</i>)	reassign to their own modules

- **Module boundary is currently wrong (too broad):** it inherits all of KEGG map04122. **Recommend a dedicated "tRNA wobble 2-thiouridine (s²U34) sulfur relay" module** scoped to the 7 core genes, explicitly excluding Moco, thiamine/s⁴U, and Fe–S branches.
- **GO curation:** MnmA → GO:0002143 (tRNA wobble position uridine thiolation) / GO:0004808; TusA/B/C/D/E → GO:0097163 (sulfur carrier activity) / GO:0002143; avoid attaching the specific thiolation GO to the Moco/thiamine genes. Request/verify TusB/TusC gene symbols in UniProt for PP_3994/PP_3995. Correct PP_3994's protein name to "sulfur relay protein TusC."

7. Genes to Promote to Full fetch-gene Review

1. **PP_4014 (mnmA)** — module anchor; confirm active-site Cys pair and tRNA specificity.
2. **PP_3993–PP_3996 (tusD/tusC/tusB/tusE cluster)** — confirm operon, fix PP_3994/PP_3995 symbols and PP_3994 name.
3. **PP_2116 (tusA)** — confirmed relay carrier (57.5% id to EcTusA, IPR022931); low priority now. **PP_1233 (tusA-I)** — assign its true branch (Moco/DsrE2-type?).

4. **PP_0842 (iscS)** — confirmed relay/isc-operon desulfurase; low priority. **PP_2435 (iscS-II)** — clarify accessory role (TA-linked desulfurase?).
-

8. Key References

- Ikeuchi, Shigi, Kato, Nishimura, Suzuki (2006) *Mol Cell*. **Mechanistic insights into sulfur relay... thiouridine at tRNA wobble**. PMID **16387657**. — Discovery of tusABCDE; in vitro reconstitution of IscS→TusA→TusBCD→TusE→MnmA.
- Numata, Fukai, Ikeuchi, Suzuki, Nureki (2006) *Nature/PNAS*. **Structural basis for sulfur relay... TusBCD complex**. PMID **16472754**. — TusBCD heterohexamer; catalytic TusD Cys78.
- Numata, Ikeuchi, Fukai, Suzuki, Nureki (2006) *Nature*. **Snapshots of tRNA sulphuration via an adenylated intermediate**. PMID **16871210**. — MnmA–tRNA structures; adenylated U34 intermediate.
- Rozov et al. (2016) *Nat Commun*. **Novel base-pairing at the tRNA wobble position**. PMID **26791911**. — Decoding role of mnm⁵s²U34.
- Yildiz, Leimkühler (2021) *J Bacteriol*. **TusA... links translation efficiency to cell division in E. coli**. PMID **33526615**. — TusA multifunctionality / sulfur branch points.

*Gene identities and lengths verified against UniProt proteome UP000000556 (Q88 accessions).**
